JS HTML DOM

Overview

JavaScript HTML DOM

-Accessing Elements

-Modifying Elements

-Manipulating Classes

-Event Handling

-Creating and Modifying Elements

JS HTML DOM

JavaScript HTML DOM (Document Object Model) is a programming interface that allows JavaScript to interact with HTML elements on a web page. It provides a structured representation of the HTML document, allowing you to access, manipulate, and modify HTML elements, attributes, styles, and content dynamically.

- **Accessing Elements:**

You can access HTML elements using various methods provided by the DOM, such as getElementById(), getElementsByClassName(), getElementsByTagName(), or querySelector().

```
// Accessing an element by its ID
const elementById = document.getElementById('myElementId');
// Accessing elements by class name
const elementsByClass = document.getElementsByClassName('myClassName');
// Accessing elements by tag name
const elementsByTag = document.getElementsByTagName('div');
// Accessing an element using CSS selector
const elementBySelector = document.querySelector('#myElementId .myClassName');
```

- **Modifying Elements:**

You can modify HTML elements by changing their attributes, styles, or content using properties and methods provided by the DOM.

```
const element = document.getElementById('myElementId');

// Changing element attributes
element.setAttribute('src', 'newimage.jpg');

// Modifying element styles
element.style.color = 'red';
element.style.fontSize = '18px';

// Changing element content
element.innerHTML = 'New content';
```

- **Manipulating Classes:**

You can add, remove, or toggle CSS classes on elements using the classList property and its associated methods.

```
const element = document.getElementById('myElementId');

// Adding a class
element.classList.add('newClass');

// Removing a class
element.classList.remove('oldClass');

// Toggling a class
element.classList.toggle('active');
```

- **Event Handling:**

You can attach event handlers to HTML elements to respond to user interactions or other events using various event listeners.

```
const button = document.getElementById('myButtonId');

// Attaching a click event handler
button.addEventListener('click', function() {
  // Code to execute when the button is clicked
});

// Removing an event handler
button.removeEventListener('click', eventHandlerFunction);
```

- **Creating and Modifying Elements:**

You can create new HTML elements dynamically and append them to the document using the createElement() method and manipulate their attributes, styles, and content.

```
const newElement = document.createElement('div');

// Modifying attributes of the new element
newElement.setAttribute('id', 'newElementId');
newElement.setAttribute('class', 'newElementClass');

// Changing content of the new element
newElement.innerHTML = 'New element content';

// Appending the new element to an existing element
const container = document.getElementById('containerId');
container.appendChild(newElement);
```

Summary

Operation	DOM Method Used
Select element	getElementById()
Change text	textContent
Change style	style.property
Add element	createElement()
Remove element	remove()
Toggle class	classList.toggle()
Handle events	addEventListener()

References

https://www.w3schools.com/js/js_htmldom.asp

https://www.w3schools.com/js/js_htmldom_document.asp

https://www.javatpoint.com/javascript-array

https://www.w3schools.com/js/js_htmldom_methods.asp

https://www.w3schools.com/js/js_htmldom_methods.asp

https://www.w3schools.com/js/js_dom_examples.asp

https://www.javatpoint.com/document-object-model